Yijingxiu (Louise) Lu

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Education

- **Seoul National University** *Sep. 2018 – Dec. 2025*
  Computer Science, Master and Ph.D. Seoul, South Korea
  - GPA: 3.83/4.20

- **University of Science and Technology Beijing** *Jun. 2017*
  Communication Engineering, Bachelor Beijing, China
  - GPA: 3.34/4.00

Professional Experience

- **Senior Algorithm Researcher** *Sep. 2026 – Present*
  Immunocan, Shanghai, China
  - Training a VHH-specific protein language model from scratch to learn nanobody sequence representations for downstream antibody discovery and engineering tasks.
  - Authoring a survey of computational methods for antibody discovery and engineering.
  - Developing and training an antibody–antigen ranking model to prioritize candidate binders.

- **Senior AI Research Scientist** *Jan. 2026 – Sep. 2026*
  Clickmab Biotechnology, Beijing, China
  - Developed antibody affinity maturation prediction models based on protein language models, achieving a Pearson correlation coefficient of 0.97 on HER2 antibody variants.
  - Evaluated and benchmarked state-of-the-art antibody–antigen complex prediction systems, including AlphaFold3, Chai-1, Protenix, and GRASP, on proprietary datasets; improved inference strategies, increasing the number of successful predictions (DockQ $>$ 0.23) from baseline performance to 26 out of 32 complexes.
  - Built machine learning models for antibody viscosity prediction, achieving generalization performance comparable to state-of-the-art published neural network approaches (test-set AUPRC = 0.85).
  - Developing computational pipelines for nanobody discovery, including positive antibody screening, antibody structure prediction, and antigen–antibody complex modeling.

- **Scientist, Pre-appointed** *Aug. 2025 – Dec. 2025*
  Global Health Drug Discovery Institute (GHDDI), Beijing, China
  - Extended the TamGen molecular foundation model by incorporating scaffold-constrained molecular generation capabilities, enabling structure-preserving lead optimization and scaffold decoration for drug discovery applications.
  - Generated approximately 270,000 target-specific small molecules across 338 malaria-related protein targets using TamGen-based generative pipelines.
  - Contributed to the development of the AI-powered drug design platform (AI KongMing), supporting molecular generation, optimization, and virtual screening workflows.

Patents and Publications

C=Conference, J=Journal, P=Patent, W=Workshop, S=In Submission

[W.4] **Lu Y.**, Li Y., Yang W., Chen X., & Wang T. (2026). **EPIC: Epitope–Paratope Interleaved Contact Learning for Antibody–Antigen Interface Prediction** *NeurIPS 2026 AI4DD Workshop*. (Accepted; forthcoming).

[J.5] Tarantino B., Kim S., **Lu Y.**, & Giudici P. (2026). **ISAAC: Prior-aligned structural sensitivity auditing for drug–target interaction models** *Artificial Intelligence in the Life Sciences*, 100180.

[W.3] Shen M., **Lu Y.**, Feng B., & King I. (2025). **Attentive Multi-Channel Molecular Representation in Drug–Target Affinity Prediction** *NeurIPS 2025 Workshop on Non-Euclidean Foundation Models and Geometric Learning: Advancing AI Beyond Euclidean Frameworks*.

[J.4] **Lu Y.**, Lee S., Kang S., & Kim S. (2026). **EnsDTI: Predicting Drug-Target Interaction with Mixture-of-Experts and Confidence Assessment** *IEEE/ACM Transactions on Computational Biology and Bioinformatics*.

[J.3] **Lu Y.**, Piao Y., Shen M., & Kim S. (2025). **Improving Reaction Yield Prediction with Chemical Atom-Level Reaction Learning**. *Journal of Chemical Information and Modeling*.

[C.4] Southiratn T., Koo B., **Lu Y.**, & Kim S. (2025). **CombiMOTS: Combinatorial Multi-Objective Tree Search For Dual-Target Molecule Generation**. In *Forty-second International Conference on Machine Learning (ICML)*.

[W.2] **Lu Y.**, Lee S., Kang S., & Kim S. (2026). **Mixture-of-Experts Approach for Enhanced Drug-Target Interaction Prediction and Confidence Assessment**. Accepted at BIOKDD 2024.

[J.2] [W.1] **Lu Y.**, Piao Y., Lee S., & Kim S. (2025). **Context-Aware Hierarchical Fusion for Drug Relational Learning**. Accepted at BIOKDD 2024. *IEEE Transactions on Computational Biology and Bioinformatics*.

[C.3] Piao Y., Lee S., **Lu Y.**, & Kim S. (2024). **Improving Out-of-Distribution Generalization in Graphs via Hierarchical Semantic Environments**. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

[C.2] **Lu Y.**, Piao Y., & Kim S. (2024). **Enhancing Drug-Drug Interaction Prediction with Context-Aware Architecture**. In *The Second Tiny Papers Track at ICLR 2024, Notable, Oral Presentation*.

[C.1] Choi M. G., Shin W., **Lu Y.**, & Kim S. (2023). **Triangular Contrastive Learning on Molecular Graphs**. In *Molecular Machine Learning Conference (MOML 2023)*.

[P.1] Lim S., Kim S., **Lu Y.**, et al. (2022). **Device and Method for Predicting Interactions Between Compounds and Proteins**. Assigned to Seoul National University, SNU-2022-06253.

[J.1] Lim S.*, **Lu Y.***, Cho C. Y., et al. (2021). **A Review on Compound–Protein Interaction Prediction Methods: Data, Format, Representation and Model**. *Computational and Structural Biotechnology Journal*.

Honors and Awards

- **Outstanding Self-financed Student** *Jun. 2025*
  The Chinese Government Award for Outstanding Self-financed Students Abroad
- **Excellent Poster Award** *Oct. 2024*
  BIOINFO/2024 Annual Conference of Korean Society for Bioinformatics
  ◦ Presented paper: "Condition Aware Relational Learning for Chemical Reaction Yields Prediction."
- **Notable Paper Award** *May 2024*
  The Second Tiny Papers Track at ICLR
  ◦ Presented paper: "Enhancing Drug-Drug Interaction Prediction with Context-Aware Architecture."
- **Samsung HumanTech Paper Award** *Feb. 2024*
  Samsung Advanced Institute of Technology / Samsung
  ◦ Presented paper: "Improving Out-of-Distribution Generalization in Graphs with Hierarchical Semantic Environments" (co-author).
- **Excellent Project Award** *Dec. 2022*
  "Chunhui Cup" Overseas Students Innovation and Entrepreneurship Competition, Ministry of Education, PRC
  ◦ Led project: "Artificial Intelligence-Based COVID-19 Target Discovery and Drug Design."
- **Global AI & Big Data Scholarship** *Jan. 2021 – Jan. 2022*
  Daewoong Foundation
  ◦ Project focus: improving drug-likeness of generated molecules using designed objective functions and a VAE framework.
- **Global Scholarship II** *Mar. 2019 – Jun. 2021*
  Office of International Affairs, Seoul National University

Teaching Experience

- **Teaching Assistant, ML-Bio** *Mar. 2025 – Jun. 2025*
  Seoul National University, Department of Computer Science
  ◦ Delivered lectures on artificial intelligence technologies and applications in bioinformatics.
- **Head Teaching Assistant, Algorithm** *Sep. 2024 – Dec. 2024*
  Seoul National University, Department of Computer Science
  ◦ Coordinated teaching assistant activities and mentored team members for the undergraduate algorithm course.
- **Teaching Assistant, Machine Learning in Bioinformatics** *Mar. 2024 – Jun. 2024*
  Seoul National University, Department of Computer Science
  ◦ Delivered lectures on deep learning methodologies and their applications in bioinformatics.
- **Teaching Assistant, AI-BIO** *Sep. 2022 – Dec. 2022*
  Seoul National University, Artificial Intelligence Institute
  ◦ Delivered lectures on AI-based drug discovery, with a focus on drug–target interaction prediction models.

Skills

- **Programming Languages:** Python, C++, MATLAB, LaTeX
- **Machine Learning & Data Science:** PyTorch, TensorFlow, HuggingFace, PyG, TorchDrug, RDKit, Matplotlib, Jax
- **Computational Biology & Drug Discovery Tools:** AutoDock Vina, ChemDraw, PyMOL, Schrödinger, Stardrop, Cytoscape
- **Web Technologies:** HTML, CSS, JavaScript, Django, D3.js, 3Dmol.js, Bootstrap
- **Database Systems:** SQL
- **Research Skills:** Experimental Design, Data Analysis, Literature Review, Interdisciplinary Collaboration

Academic Service

KDD 2025 Research Track Excellent Reviewer.
KDD 2024 Research Track Commendable Reviewer.

Languages

Chinese: Native | **English**: Business and Academic Proficiency | **Korean**: Business Proficiency